Marwadi University Marwadi Chandarana Group NAAC A+	**Marwadi University** **Faculty of Engineering & Technology** **Department of Information and Communication Technology**	
Subject: Digital Signal and	**Aim: Perform gray level operations images.**	
Experiment No: 7		92400133104

Aim: Perform gray level operations images.

Theory:

Gray level operations involve manipulating the pixel values of an image to enhance or modify its appearance. These operations are commonly used in image processing tasks such as contrast adjustment, brightness correction, and image thresholding.

Flowchart:

The flowchart for the gray level operations program will consist of the following steps:

Load the input image.
Convert the image to grayscale.
Perform the desired gray level operation.
Display the processed image.
Save the processed image (optional).

```
import cv2
import numpy as np
import matplotlib.pyplot as plt

threshold_val = 30
gamma_val = 2

src_image = cv2.imread('/content/pexels-steve-29833481.jpg',
cv2.IMREAD_GRAYSCALE)

if src_image is None:
  print("Image not found! check file path.")
  exit()
negative_image = 255 - src_image

_, thresholded_image = cv2.threshold(src_image, threshold_val, 255,
cv2.THRESH_BINARY)
```

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```
invGamma = gamma_val
table = np.array([((i / 255.0) ** invGamma) * 255 for i in np.arange(0,
256)]).astype("uint8")
gamma_corrected_image = cv2.LUT(src_image, table)

plt.figure(figsize=(12, 12))

plt.subplot(1, 4, 1)
plt.imshow(src_image, cmap='gray', vmin = 0, vmax = 255)
plt.title('Original Image', fontweight = 'bold')
plt.axis('off')

plt.subplot(1, 4, 2)
plt.imshow(negative_image, cmap='gray', vmin = 0, vmax = 255)
plt.title('Negative Image', fontweight = 'bold')
plt.axis('off')

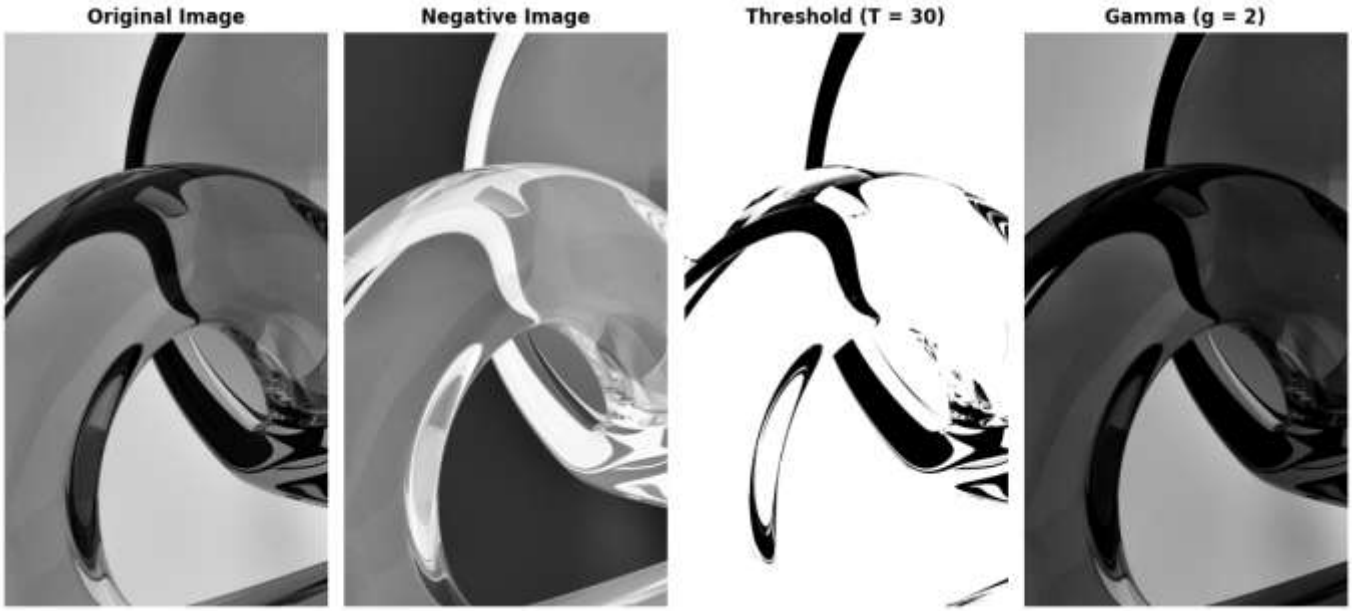
plt.subplot(1, 4, 3)
plt.imshow(thresholded_image, cmap='gray', vmin = 0, vmax = 255)
plt.title(f"Threshold (T = {threshold_val})", fontweight = 'bold')
plt.axis('off')

plt.subplot(1, 4, 4)
plt.imshow(gamma_corrected_image, cmap='gray', vmin = 0, vmax = 255)
plt.title(f"Gamma (g = {gamma_val})" , fontweight = 'bold')
plt.axis('off')

plt.tight_layout()
plt.show()
```

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Gama = 2 Threshold = 30

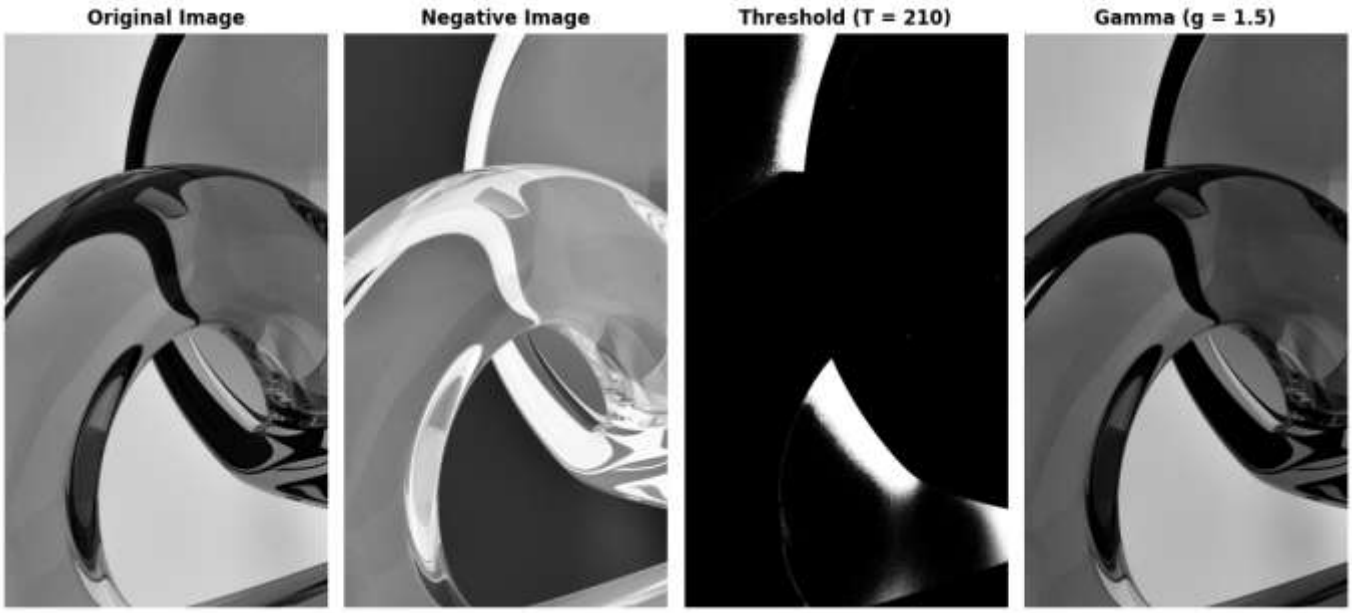


Low threshold makes most. pixels white, while high gamma darkens the image before/around the processing. Important grey details are poorly represented.

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Gama = 1.5 Threshold = 210

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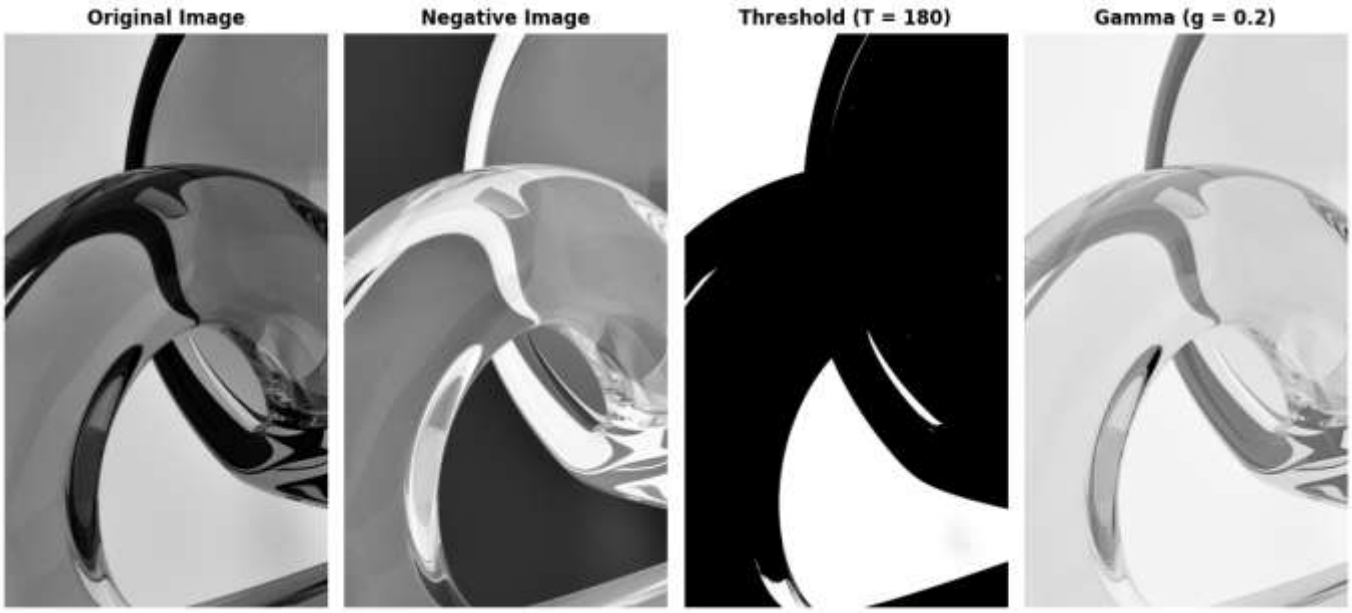


Very high threshold removes most of the image into black, and gamma > 1 further darkens the grey regions.

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Gama = 0.2 Threshold = 180

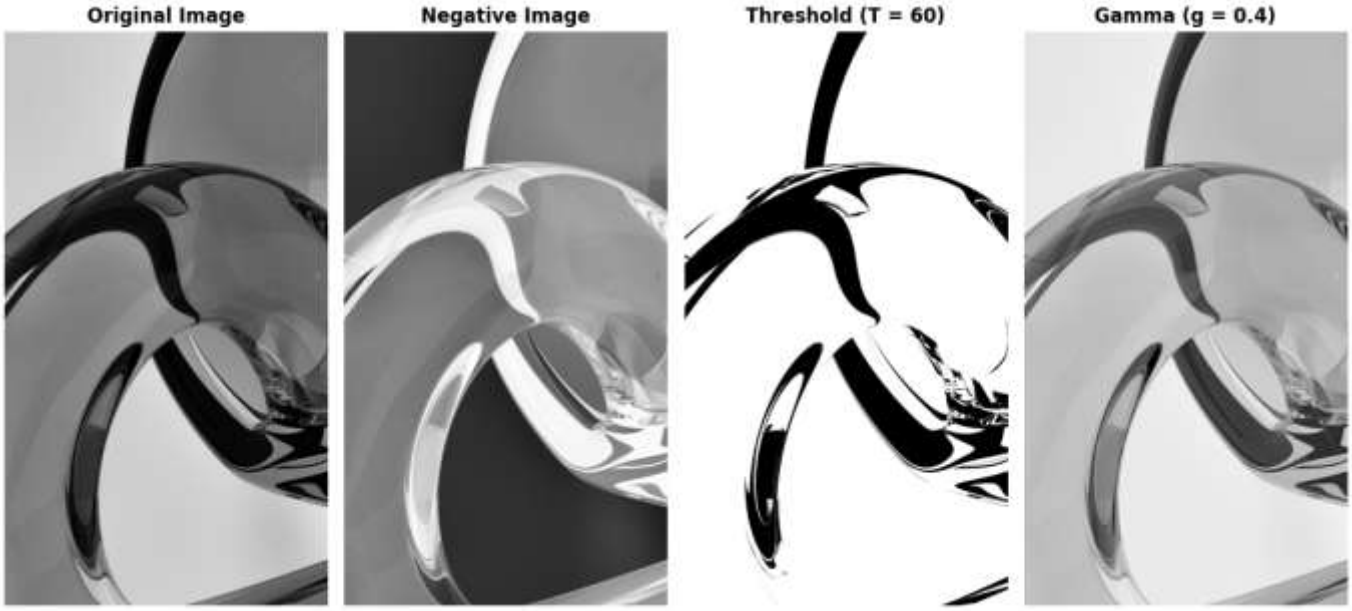


Gamma < 1 brightens the image, but the very high threshold still removes most medium-grey regions.

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Gama = 0.4 Threshold = 60

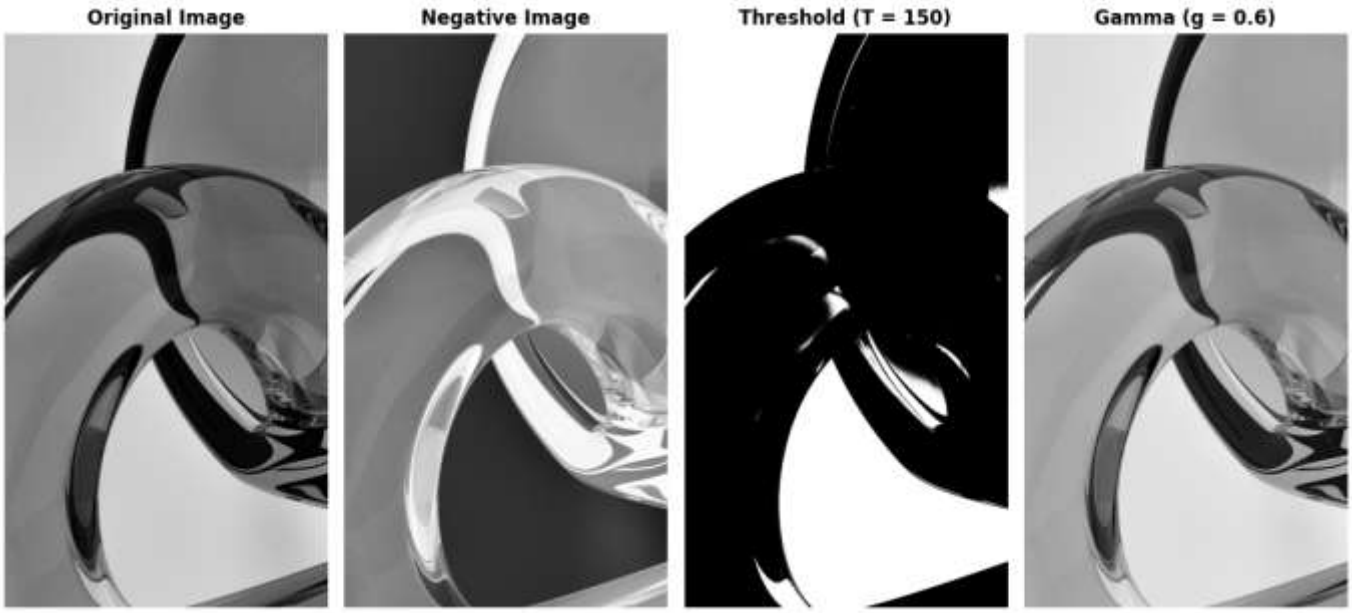


Low threshold retains many white pixels and gamma < 1 brightens the darker curved regions. Some contrast is lost.

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Gama =0.6 Threshold = 150

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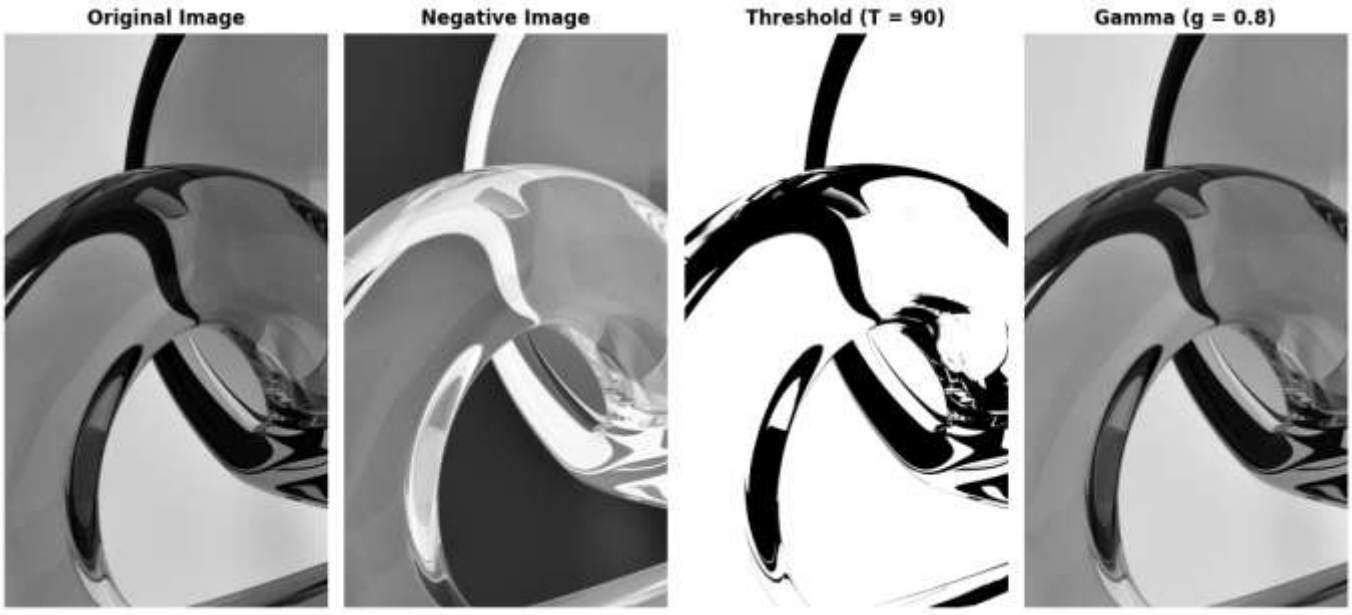


Gamma 0.6 brightens darker details while threshold 150 separates much of the dark structure from the bright background.

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Gama = 0.8 Threshold = 90

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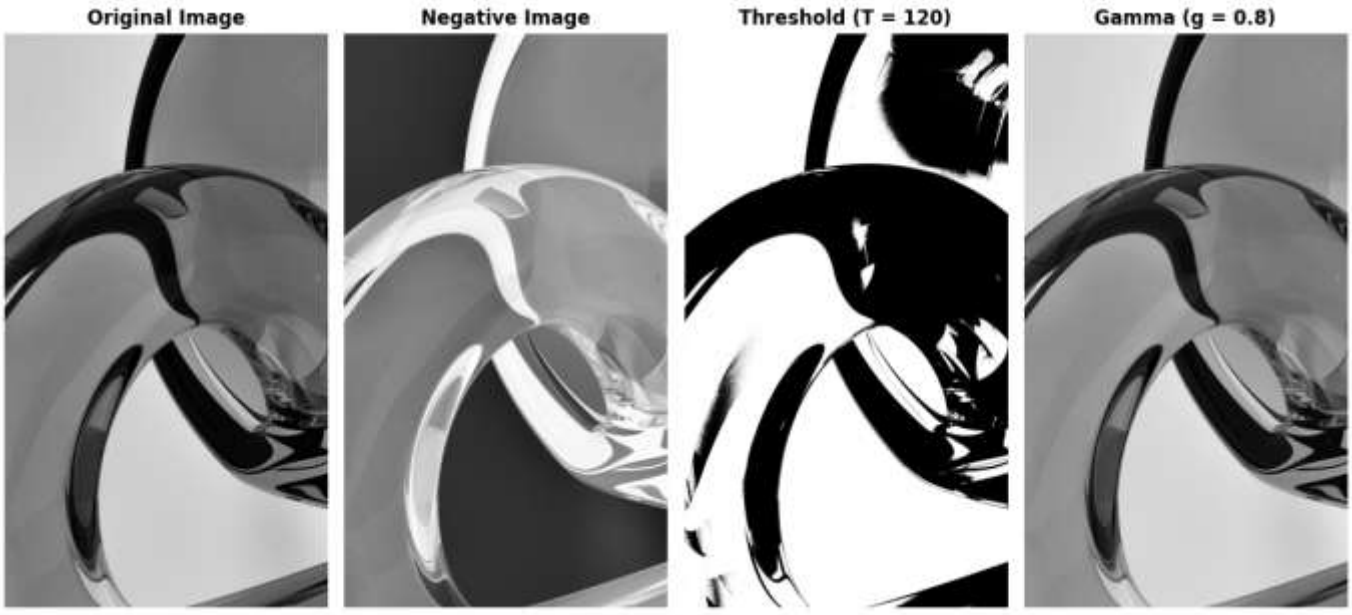
Gamma 0.8 slightly brightens the image and threshold 90 gives good separation between the curved dark object and the background.

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Best output:

Gama =0.8 Threshold = 120

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Threshold 120 provides strong object/background separation while gamma 0.8 slightly improves visibility of grey regions without excessive brightening.

Conclusion :

The combination of gamma correction and thresholding produces different segmentation results depending on both parameters. Low gamma values brighten darker regions, while high gamma values darken them. A low threshold produces more white pixels, whereas a high threshold produces more black pixels. Extreme combinations such as $T = 30$, $\gamma = 2.0$ or $T = 210$, $\gamma = 1.5$ result in significant loss of image details. The combination $T = 120$ and $\gamma = 0.8$ provides a better balance between brightness and binary segmentation for the given image, preserving the major curved structures while maintaining separation from the bright background.